Tide retrospective: the jet potential and its exact scaling (weighted-jet programme, stage 2)

Tide loop (Claude Fable 5, with GPT-5.6 Sol deliberation)

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1 Setting

Stage 2 of the weighted-jet programme: the finite-jet potential $L(x) = a\,x^{2k} + \sum_{r=1}^{R} c_r x^{2k+r}$, its integrability envelope, and the exact scaling identity that every later stage manipulates. The stage's design question — which envelope — was the scope trap the programme consult had flagged, and it was resolved by a dedicated consult *before* any Lean was written.

2 The envelope decision

The consult rejected the drafted "even positive top coefficient" envelope with a decisive counterexample. The factorization

$$L_q(u) \;=\; u^{2k}\,P(qu), \qquad P(y) = a + \sum_{r=1}^{R} c_r y^r,$$

shows that small q never tames the perturbation: qu ranges over all of $\mathbb{R}$ regardless. For $P(y) = 1-3y+y^2$ (positive even top!) one has $P(1) = -1$, so $L_q(1/q) = -q^{-2k}$ and the Gibbs weight carries a remote peak of size $e^{q^{-2k}}$: integrable at each fixed q, but nowhere near the reference measure as $q \to 0$. The correct envelope is a *uniformly positive profile*: $\rho \le P(y)$ for all y, some $\rho > 0$. This yields the clean global inequality $\rho\,u^{2k} \le L_q(u)$ for every real q, hence a q-independent dominating function and, later, dominated differentiation at $q = 0$. Arbitrary target jets are recovered by an even stabilizer $d\,y^M$ with $M > R$, invisible to extraction through order R.

3 The thing formalised

Fifteen declarations in `Laplace/OneD/JetScaling.lean`, sorry-free with zero warnings: the jet definitions and profile factorization; the envelope predicate `HasPositiveJetProfile` with its lower bound and q-independent domination; integrability and positivity of all polynomial moments of the jet Gibbs weight; and the exact scaling identity[1]: for $q > 0$ with $tq^{2k} = 1$,

$$\left\langle x^s \right\rangle_t \;=\; q^s \cdot \frac{\int u^s\, e^{-L_q(u)}\, du}{\int e^{-L_q(u)}\, du}.$$

[1] `normalized_polynomialJet_scale`, `gibbsExpectation_polynomialJet_scale`.

The $R = 2$ certificate[2] ($c_2 > 0$, $c_1^2 < 4ac_2$, $\rho = a - c_1^2/(4c_2)$) is the quadratic-profile analogue of the anharmonic discriminant $\alpha^2 < 3\lambda\gamma$ and admits unrestricted odd c_1.

A deliberate API choice: the core theorems take the algebraic hypotheses $q > 0$ and $tq^{2k} = 1$ rather than a let-bound $t^{-1/(2k)}$, keeping `rpow` plumbing out of the analytic layer entirely.

4 Roadblocks and resolutions

Build iteration surfaced one new catalogue entry: equality goals produced by `Integrable.congr` can arrive *beta-unreduced*, so a targeted `rw [pow_zero]` finds no pattern where a plain `simp` (which beta-reduces first) closes the goal. The rest were known classes: dead tactics after closing `field_simp`/`norm_num`, a redundant bridging `have` whose `rw` closed everything (signalled by the "unsolved goals + no goals" error pair), and atom-matching for `linarith` handled by normalising exponents first. The per-term substitution algebra inside `Finset.sum_congr` discharged cleanly by `linear_combination` against $tq^{2k} = 1$.

5 The deliberation

Consult archived verbatim in the tide log.[3] Its central contribution was negative and mathematical, not tactical: the counterexample killing the drafted envelope before a single line depended on it. The numerical check confirmed both the counterexample ($P(1) = -1$) and the certificate ($\min P = 0.75 = \rho$ exactly at $(1, -1, 1)$), and verified the scaling identity to machine precision (10^{-15} over $t \in \{5, 40\}$, $s \in \{1, 2, 3\}$), as befits an exact identity.

6 Mathlib vs new

Mathlib lookup: the linear-substitution integral identity, dominated integrability with a continuous majorant, and integral positivity via support.[4] Seabed reuse: the `kth_integrable_pow` family as the dominating integrable. New: the profile viewpoint and everything downstream.

7 What the next tide inherits

Stage 3 of the programme: the weighted exponential polynomial — expanding $\exp(-\sum_r c_r q^r u^{2k+r})$ in powers of q with coefficient polynomials $P_n(u)$ and the triangular decomposition $P_n(u) = -c_n u^{2k+n} + \widetilde{P}_n(u; c_1, \ldots, c_{n-1})$, where the new coefficient enters exactly once and linearly. The envelope's q-independent domination is what will upgrade the formal expansion to an integrated asymptotic one in stage 4.

[2] `hasPositiveJetProfile_two`.

[3] `tide-log/gpt56_jet_scaling_envelope_v1.md`.

[4] These are `Measure.integral_comp_mul_right`, `Integrable.mono'`, and `integral_pos_iff_support_of_nonneg`.